Practice workbook · Finding the equation of a tangent

Finding the gradient at a point by differentiating and writing the tangent's equation, sometimes then finding where it meets an axis or another line.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-04

1. Question 8 · part b · 5 marks

2020 NOV · Paper 02

Worked-example retry · Attempt part b only.

8 The curve C_1 has equation $y = 5e^{-2x} + 4$

The curve C_2 has equation $y = e^{2x}$

The curves C_1 and C_2 intersect at the point A.

(a) Find the exact coordinates of A.

(4)

The tangent at A to C_1 intersects the x-axis at the point B.

(b) Show that the x coordinate of B is $\frac{1}{2}(5 + \ln 5)$

(5)

The tangent at A to C_2 intersects the x-axis at the point D.

(c) Find the area of ΔABD.

(6)

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1. Question 8 · part b · 5 marks · Working and answer

Working and answer

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2. Question 6 · part b · 6 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part b only.

6

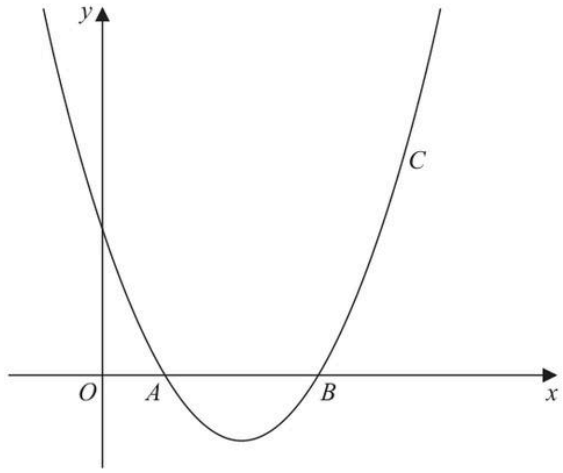


Diagram **NOT** accurately drawn

Figure 3

The curve C with equation $y = x^2 - 5x + 4$ crosses the x-axis at the points A and B, as shown in Figure 3

(a) Find the coordinates of A and the coordinates of B.

(3)

The tangent to C at A meets the tangent to C at B at the point T.

(b) Find the coordinates of T.

(6)

The normal to C at A meets the normal to C at B at the point N.

(c) Find the coordinates of N.

(3)

(d) Find the area of the quadrilateral $ATBN$.

(3)

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2. Question 6 · part b · 6 marks · Working and answer

Working and answer

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3. Question 6 · part a · 5 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

6 The curve C has equation

$$y = 4\sqrt{x}$$

The point A on C has coordinates (9, 12)

The tangent to C at the point A meets the x-axis at the point T

(a) Find the coordinates of T

(5)

The normal to C at the point A meets the x-axis at the point N

(b) Find the coordinates of N

(4)

(c) Calculate the area of triangle ATN

(2)

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3. Question 6 · part a · 5 marks · Working and answer

Working and answer

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4. Question 8 · part c · 6 marks

2020 NOV · Paper 02

Further practice · Attempt part c only.

8 The curve C_1 has equation $y = 5e^{-2x} + 4$

The curve C_2 has equation $y = e^{2x}$

The curves C_1 and C_2 intersect at the point A.

(a) Find the exact coordinates of A.

(4)

The tangent at A to C_1 intersects the x-axis at the point B.

(b) Show that the x coordinate of B is $\frac{1}{2}(5 + \ln 5)$

(5)

The tangent at A to C_2 intersects the x-axis at the point D.

(c) Find the area of ΔABD.

(6)

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4. Question 8 · part c · 6 marks

2020 NOV · Paper 02

Further practice · Attempt part c only.

Question 8 continued

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4. Question 8 · part c · 6 marks

2020 NOV · Paper 02

Further practice · Attempt part c only.

Question 8 continued

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4. Question 8 · part c · 6 marks

2020 NOV · Paper 02

Further practice · Attempt part c only.

Question 8 continued

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(Total for Question 8 is 15 marks)

4. Question 8 · part c · 6 marks · Working and answer

Working and answer

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5. Question 4 · part whole · 7 marks

2023 JAN · Paper 02

Further practice · Attempt part whole only.

4 The equation of a curve is $y = x^3 \sin x$

Find an equation of the tangent to the curve at the point on the curve where $x = \frac{1}{2}\pi$

Give your answer in the form $y = mx + c$

(7)

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5. Question 4 · part whole · 7 marks

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Further practice · Attempt part whole only.

Question 4 continued

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(Total for Question 4 is 7 marks)

5. Question 4 · part whole · 7 marks · Working and answer

Working and answer

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6. Question 3 · part b · 3 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

3 Given that $y = e^{2x}(x^2 + 1)$

(a) find $\frac{dy}{dx}$

(3)

The straight line l is the tangent to the curve with equation $y = e^{2x}(x^2 + 1)$ at the point on the curve where $x = 0$

(b) Find an equation for l in the form $y = mx + c$

(3)

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6. Question 3 · part b · 3 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 3 continued

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(Total for Question 3 is 6 marks)

6. Question 3 · part b · 3 marks · Working and answer

Working and answer

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7. Question 10 · part a · 5 marks

2025 NOV · Paper 02

Further practice · Attempt part a only.

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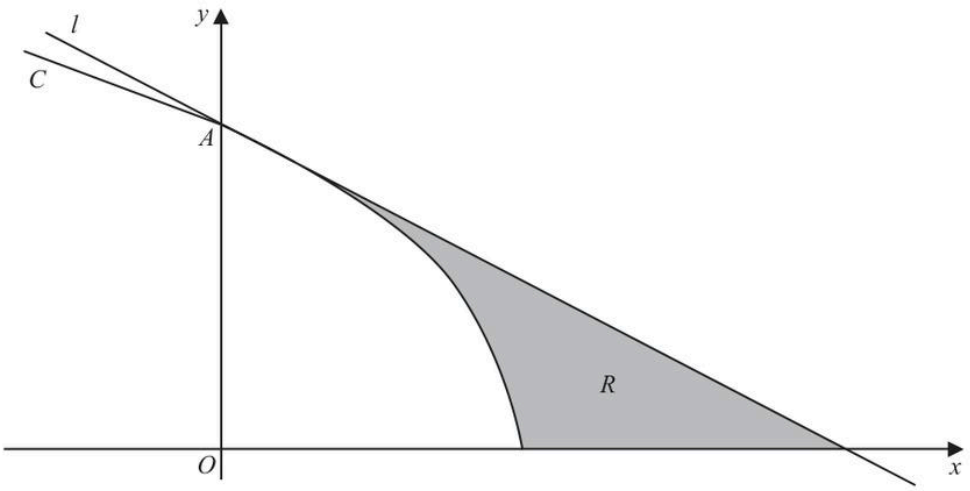


Figure 3

Figure 3 shows part of the curve C with equation $y = 3\sqrt{4-x}$ and part of the line l with equation $px + qy + r = 0$ where p and r are integers and q is prime.

The curve C cuts the y-axis at the point A

The line l is the tangent to C at A

(a) Find the value of p, the value of q and the value of r

(5)

The finite region R, shown shaded in Figure 3, is rotated through 360° about the x-axis.

(b) Use algebraic integration to find the volume of the solid formed.
Give your answer in terms of π

(5)

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7. Question 10 · part a · 5 marks · Working and answer

Working and answer

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8. Question 10 · part c · 4 marks

2021 NOV · Paper 01

Further practice · Attempt part c only.

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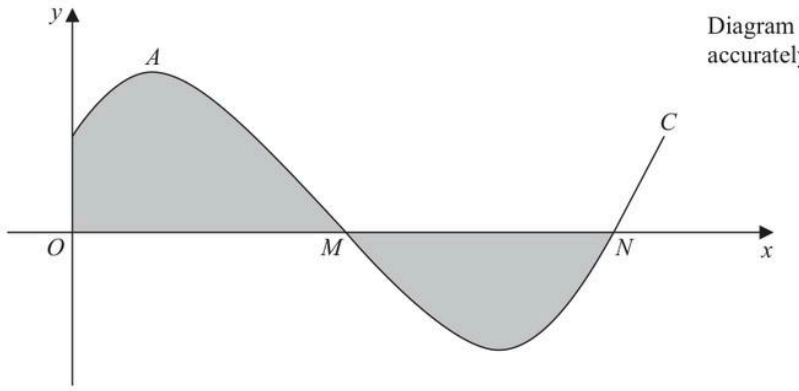


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows the curve C with equation $y = \frac{1}{2} + \sin 3x$ where $0 \leqslant x < \frac{2\pi}{3}$

The curve C crosses the x-axis at the points M and N

(a) Show that the coordinates of M are $\left(\frac{7\pi}{18}, 0\right)$ and find the coordinates of N

(3)

The curve C has a maximum at the point A

(b) Find the coordinates of A

(4)

(c) Find an equation of the tangent to C at M

Give your answer in the form $ay + b\sqrt{3}x - c\sqrt{3}\pi = 0$ where a, b and c are integers to be found.

(4)

The finite region, shown shaded in Figure 4, is bounded by the curve C, the y-axis and the part of the x-axis from O to N

(d) Use algebraic integration to find, to 3 significant figures, the total area of the shaded region.

(4)

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8. Question 10 · part c · 4 marks · Working and answer

Working and answer



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9. Question 6 · part b · 5 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

6 The curve C has equation $y = \dfrac{e^{(x^2+1)}}{x^2+1}$

(a) Show that $\dfrac{dy}{dx} = \dfrac{Kx^3e^{(x^2+1)}}{(x^2+1)^2}$ where K is a constant whose value is to be found.

(5)

(b) Find an equation of the tangent to C at the point on C where $x = -1$
Simplify your answer.

(5)

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9. Question 6 · part b · 5 marks

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Further practice · Attempt part b only.

Question 6 continued

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(Total for Question 6 is 10 marks)

9. Question 6 · part b · 5 marks · Working and answer

Working and answer

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